

Question			Marking details	Marks available					
				AO1	AO2	AO3	Total	Maths	Prac
		(iii)	F (for 24 m s^{-1}) = 23.5 [N] ecf (1) $\Sigma F = 23.5$ ecf – 20 (or candidate answer to (b)(ii)) = 3.5 [N] (or 3.7 N) (1) Mass determined = 2.0 k[g] (1) $a = \frac{\Sigma F}{m} = \frac{3.5}{2.0 \text{ ecf}} = 1.75 \text{ m s}^{-2}$ or 1.85 m s^{-2} (1) unit mark Accept correct answer based on candidate figures OR: Force = 23.5 [N] (obtaining force) (1) Mass = 2.015 k[g] (obtaining mass) (1) $a = \frac{\Sigma F}{m} = \frac{23.5}{2.015} = 11.67$ and subtracting acceleration due to gravity (1) Answer = 1.85 m s^{-2} (1) unit mark	1	1 1 1		4	4	
	(c)	(i)	Rate of change of displacement Accept $\text{velocity} = \frac{\text{displacement}}{\text{time}}$ Don't accept distance in a given direction $\div$ time or displacement in a given period of time	1			1		